Intro to ML and k nearest neighbours

Machine Learning, lecture 1

Makhmudjon Sodikov
YMLS Uzbekistan, fall 2026

YMLS



girafe ai

Today

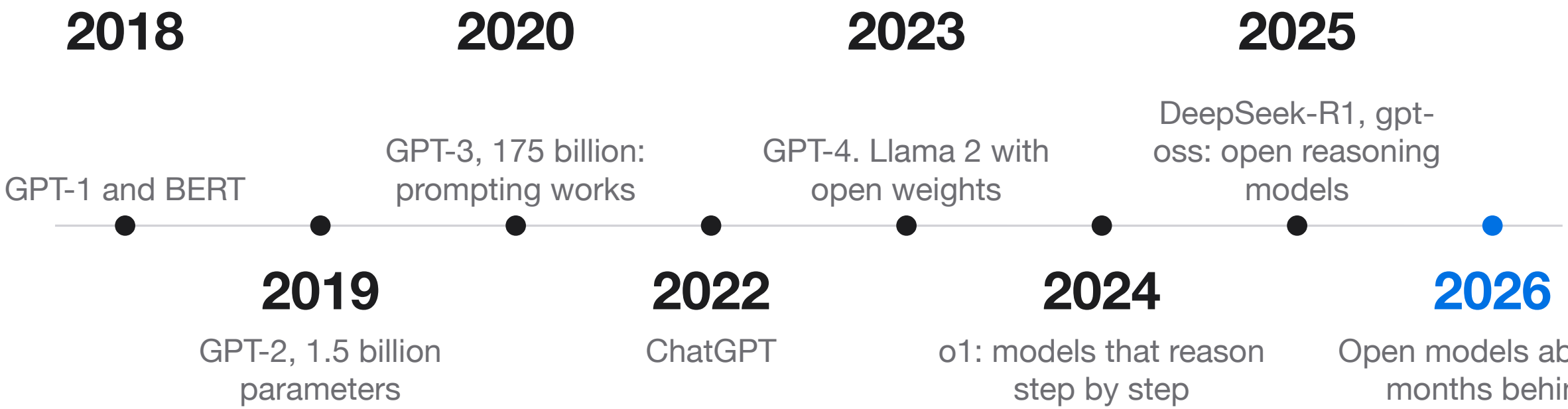
1. What machine learning is, and its vocabulary
2. Data: where to find it, how to explore it
3. How an ML project works
4. Our first model: k nearest neighbours

Machine learning today

A short history of AI



Language models, 2018 to 2026



Benchmarks fill up fast

Benchmark	What it tests	Best at launch	Best, Nov 2026
MMLU, 2020	57 school and university subjects	44%	**90%+**
Humanity's Last Exam, 2025	2,500 expert questions	8%	**61%**
ARC-AGI-2, 2025	Puzzles that people find easy	under 10%	**95%**
ARC-AGI-3, 2026	Interactive puzzle games	under 1%	**63%**

Sources: Hendrycks et al. 2020, lastexam.ai, Artificial Analysis, arcprize.org

Vision: detect, segment, generate



Games: AlphaGo and OpenAI Five



AlphaGo beat Lee Sedol 4 games to 1, 2016



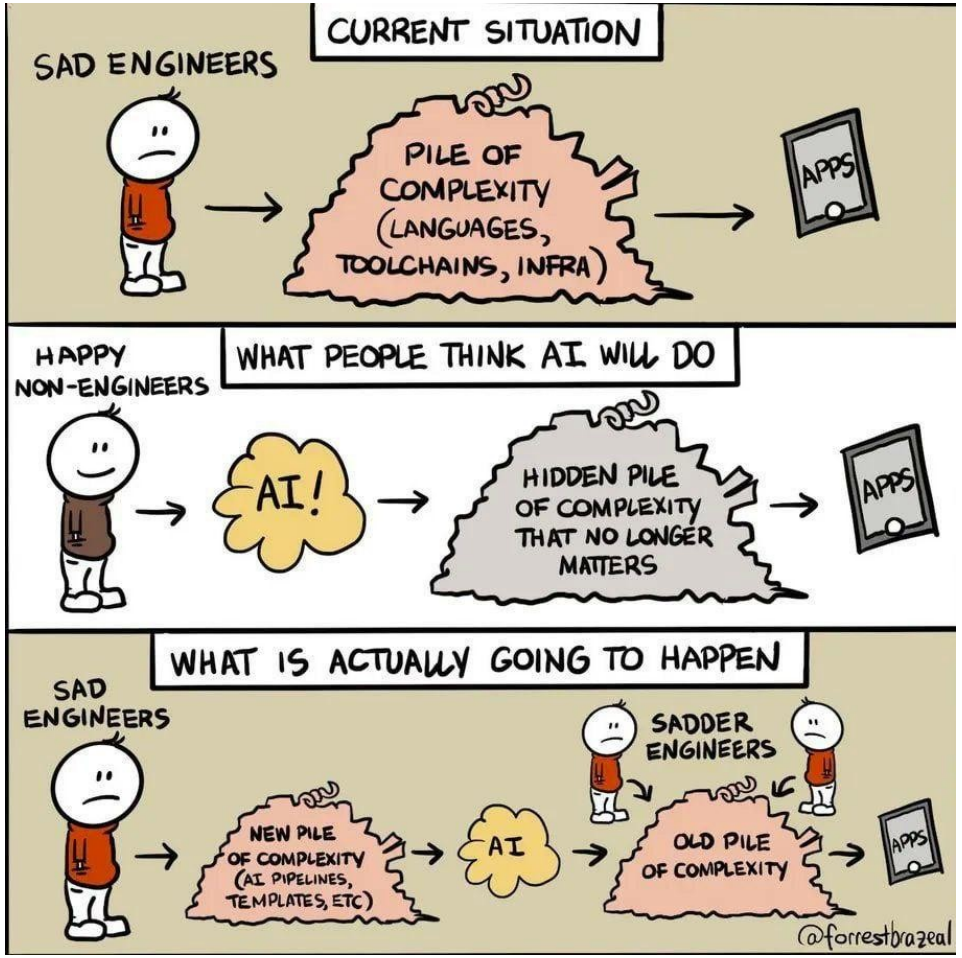
OpenAI Five beat the Dota 2 world champions, 2019

Two Nobel Prizes went to machine learning in 2024

Physics: Hopfield and Hinton, neural networks.

Chemistry: Baker, Hassabis and Jumper, protein design and AlphaFold.

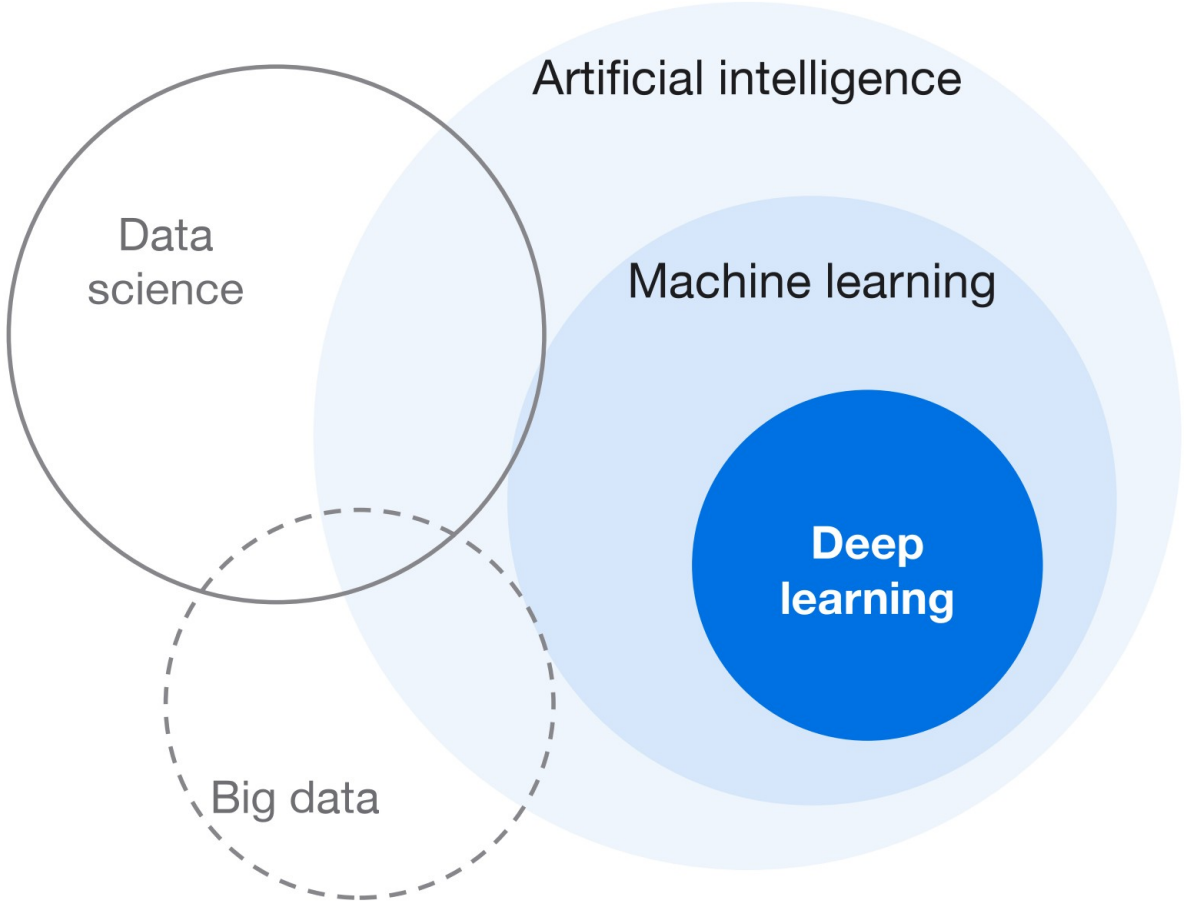
What are AI and ML for you?



Comic by Forrest Brazeal

What is machine learning?

Terms around ML



Artificial intelligence

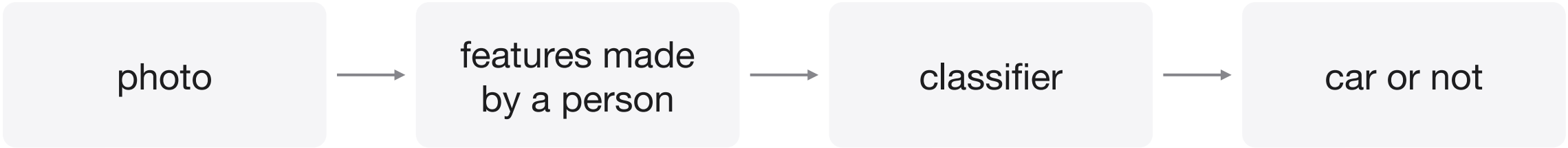


Any system that perceives its environment and acts to reach its goals.

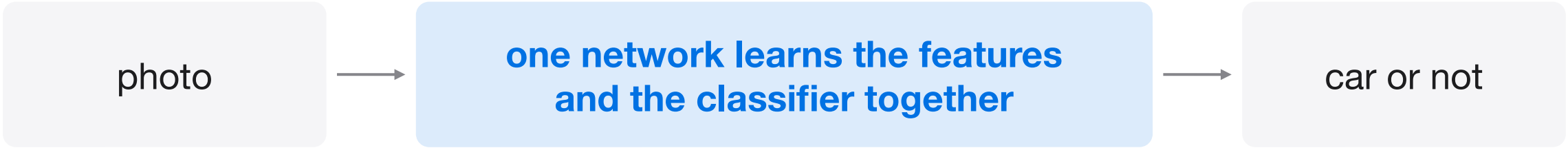
By this definition, your electric kettle is AI.

Deep learning learns the features too

Machine learning



Deep learning



Data science and big data

Data science

Statistics, programming and knowledge of the field together

The goal: insights and decisions from data

Big data

Data too large or too complex for ordinary tools

World storage: 2.6 exabytes in 1986, about 295 in 2007

Storage estimates: Hilbert and López, Science, 2011

Software 1.0, 2.0, 3.0

1.0

code

A person writes the rules

2.0

weights

Rules are learned from data and answers

3.0

prompts

Plain-language instructions for an LLM

Terms by Andrej Karpathy, 2017 and 2025

Is a review positive? Three ways

1.0 rules in code

```
GOOD = {"great", "love"}
BAD = {"boring", "awful"}

s = 0
for w in review.split():
    if w in GOOD: s += 1
    if w in BAD: s -= 1
positive = s >= 0
```

2.0 weights

10,000 good reviews
10,000 bad reviews

↓

train a classifier

↓

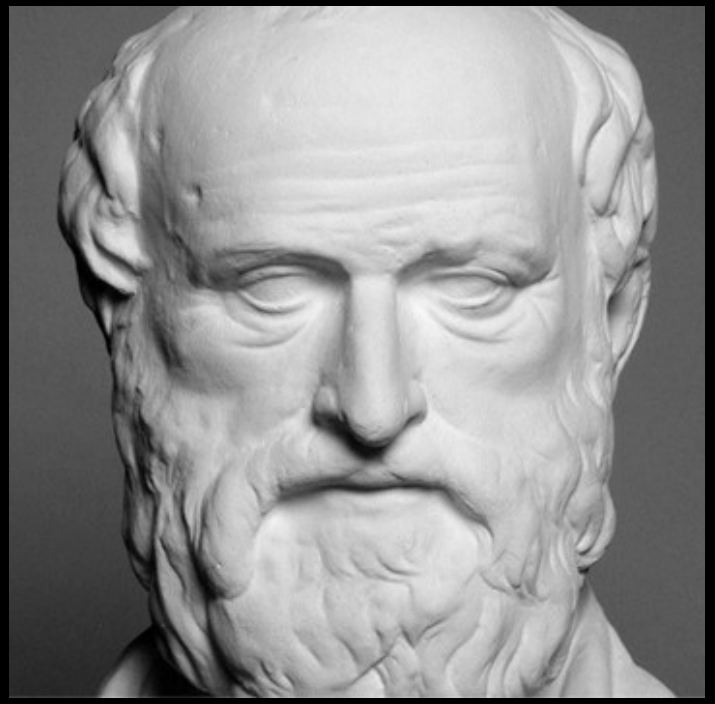
weights

3.0 a prompt

```
Reply POSITIVE or NEGATIVE.

Review: I loved this film.
POSITIVE
Review: A boring plot.
NEGATIVE
Review: {new review}
```

Long before machine learning



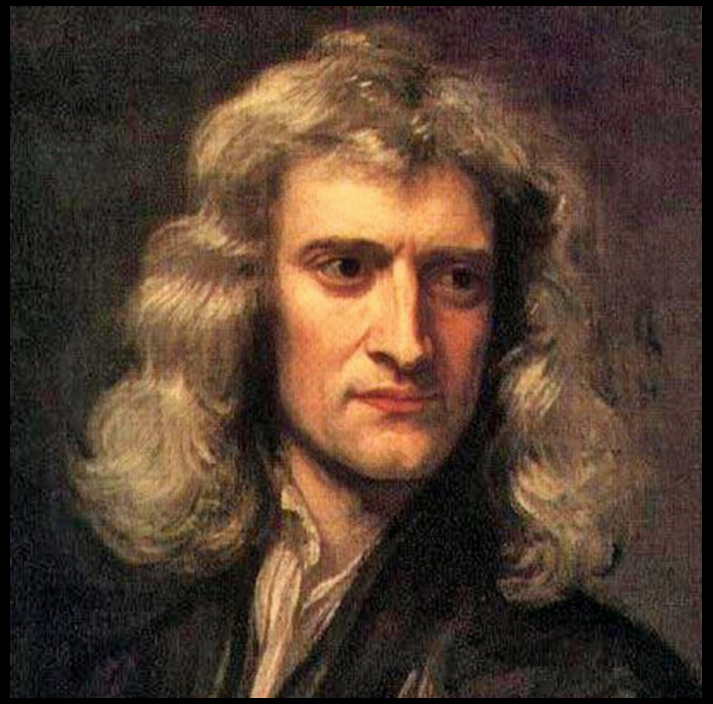
Eratosthenes

measured the Earth, about 240 BC



Johannes Kepler

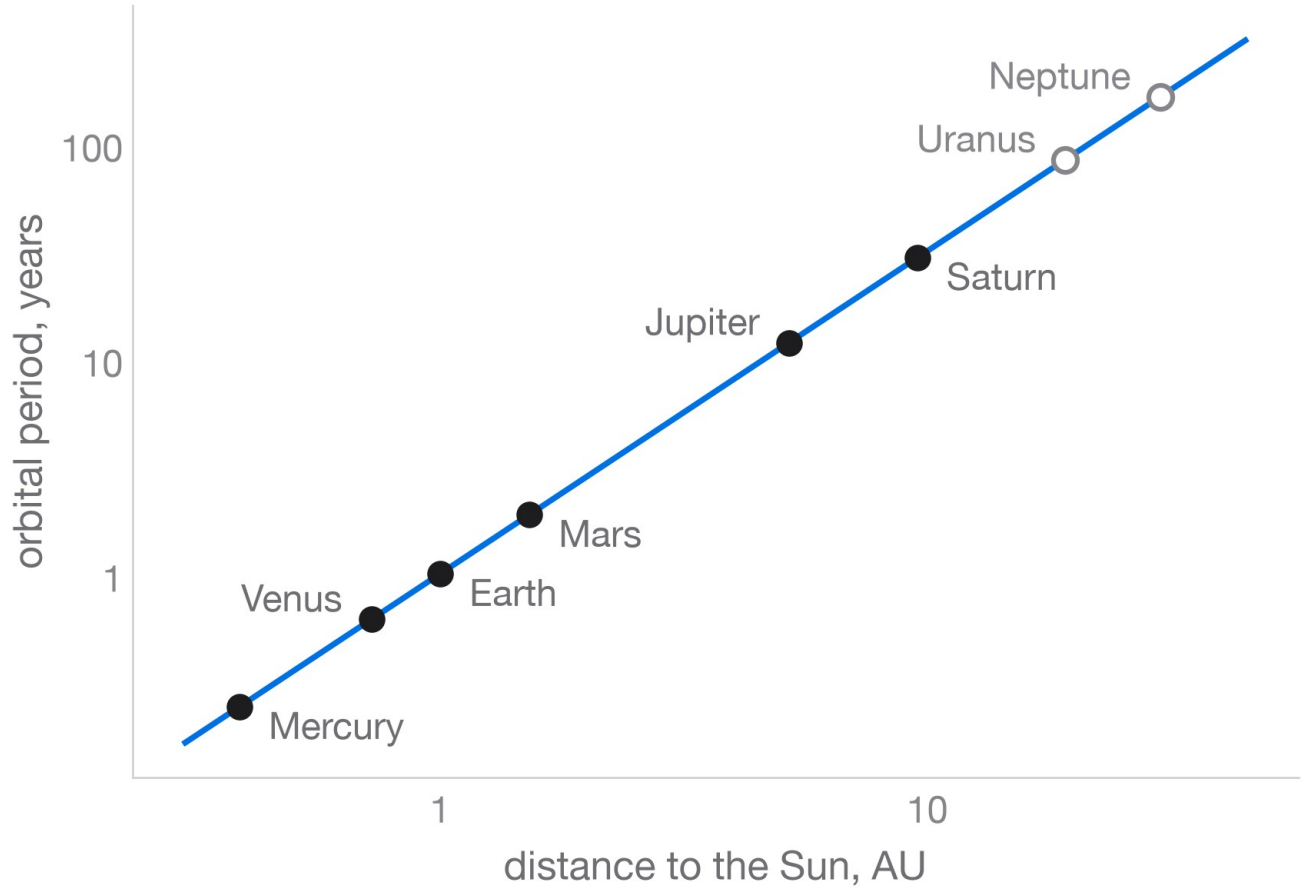
third law of planets, 1619



Isaac Newton

law of gravity, 1687

Kepler's third law, from data



$$T = C\, a^p$$

p = 1.50

fitted on the six planets Kepler knew. Uranus and Neptune, found later, fit too.

Where math ends and ML begins

1

Theory

Newton derives the law. Data only checks it.

2

Statistics

We pick the form of the law. Data picks its parameter.

3

Machine learning

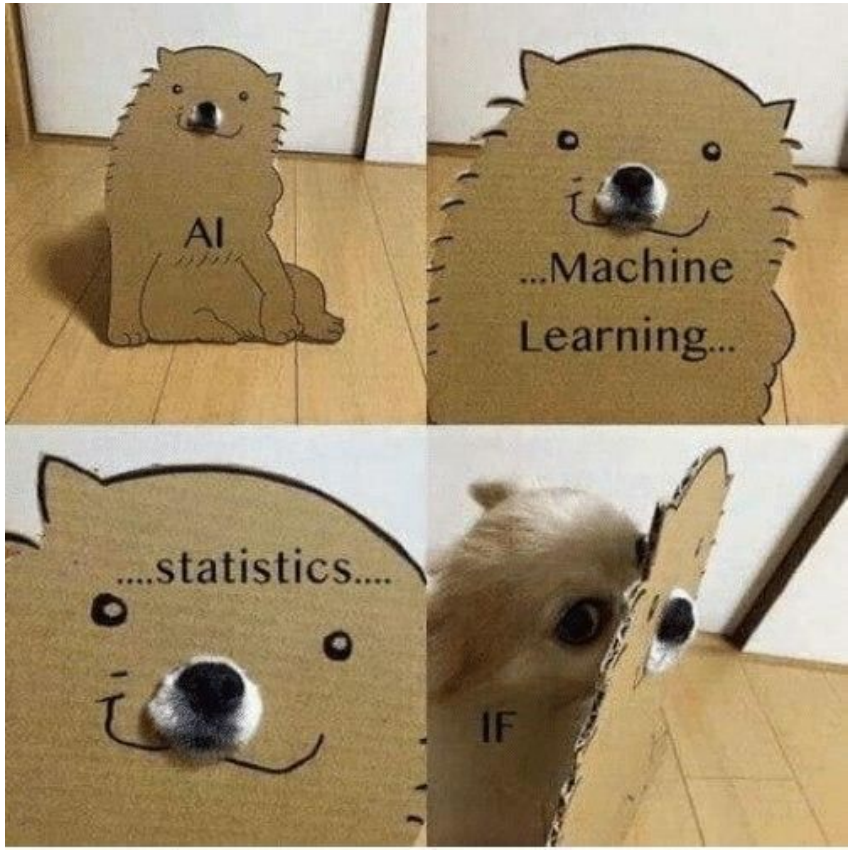
The form is unknown. We judge a flexible model by its error on new data.

4

Software 2.0

The whole program is learned.

What it actually is



IF IF IF IF IF IF IF IF IF WE!

ML thesaurus

A small task: predict the ML mark

Five students: their marks in statistics and Python, and what they got in machine learning.

name	age	statistics	python	eye colour	native language	ML mark	passed
Aziz	22	5	4	brown	Uzbek	5	yes
Malika	17	4	5	brown	Russian	4	yes
Timur	25	5	5	blue	Uzbek	5	yes
Dilnoza	27	3	4	green	Tajik	5	yes
Sardor	23	3	3	NA	Karakalpak	2	no

A new student has 4 in statistics and 5 in Python. What will they get in ML?

One row is one object

Also called an observation or a data point.

name	age	statistics	python	eye colour	native language	ML mark	passed
Aziz	22	5	4	brown	Uzbek	5	yes
Malika	17	4	5	brown	Russian	4	yes
Timur	25	5	5	blue	Uzbek	5	yes
Dilnoza	27	3	4	green	Tajik	5	yes
Sardor	23	3	3	NA	Karakalpak	2	no

We usually assume the rows are i.i.d.: independent, and drawn from the same distribution.

One column is one feature

Also called a predictor or a field.

name	age	statistics	python	eye colour	native language	ML mark	passed
Aziz	22	5	4	brown	Uzbek	5	yes
Malika	17	4	5	brown	Russian	4	yes
Timur	25	5	5	blue	Uzbek	5	yes
Dilnoza	27	3	4	green	Tajik	5	yes
Sardor	23	3	3	NA	Karakalpak	2	no

Age is a feature: one property, measured for every student.

Features come in different types

Numbers, categories, and even text.

name	age	statistics	python	eye colour	native language	ML mark	passed
Aziz	22	5	4	brown	Uzbek	5	yes
Malika	17	4	5	brown	Russian	4	yes
Timur	25	5	5	blue	Uzbek	5	yes
Dilnoza	27	3	4	green	Tajik	5	yes
Sardor	23	3	3	NA	Karakalpak	2	no

numbers categories text: even the name is a feature

The design matrix holds all features

Rows are objects, columns are features. The target stays outside.

name	age	statistics	python	eye colour	native language	ML mark	passed
Aziz	22	5	4	brown	Uzbek	5	yes
Malika	17	4	5	brown	Russian	4	yes
Timur	25	5	5	blue	Uzbek	5	yes
Dilnoza	27	3	4	green	Tajik	5	yes
Sardor	23	3	3	NA	Karakalpak	2	no

$X \in \mathbb{R}^{n \times p}$ *n* objects, *p* features. Here *n* = 5 and *p* = 6.

The target is what we predict

Also called the label or the answer.

name	age	statistics	python	eye colour	native language	ML mark	passed
Aziz	22	5	4	brown	Uzbek	5	yes
Malika	17	4	5	brown	Russian	4	yes
Timur	25	5	5	blue	Uzbek	5	yes
Dilnoza	27	3	4	green	Tajik	5	yes
Sardor	23	3	3	NA	Karakalpak	2	no

ML mark is a number: regression. **Passed** is a label: classification. $y \in \mathbb{R}^n$

A model makes predictions

Our first guess: the average of the two marks.

name	statistics	python	ML mark	prediction
Aziz	5	4	5	4.5
Malika	4	5	4	4.5
Timur	5	5	5	5
Dilnoza	3	4	5	3.5
Sardor	3	3	2	3

$$\hat{y} = \frac{1}{2}\, x_{\text{stats}} + \frac{1}{2}\, x_{\text{python}}$$

Even a random rule is a model

Different models give different predictions.

name	statistics	python	ML mark	prediction
Aziz	5	4	5	1
Malika	4	5	4	5
Timur	5	5	5	2
Dilnoza	3	4	5	4
Sardor	3	3	2	3

$$\hat{y} = \text{a random integer from 1 to 5}$$

A loss measures the error

Mean squared error of the random model.

name	ML mark	prediction	squared error
Aziz	5	1	**16**
Malika	4	5	**1**
Timur	5	2	**9**
Dilnoza	5	4	**1**
Sardor	2	3	**1**

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

$$= \frac{16 + 1 + 9 + 1 + 1}{5} = 5.6$$

Another loss: mean absolute error

The same predictions, a different way to count the error.

name	ML mark	prediction	absolute error
Aziz	5	1	**4**
Malika	4	5	**1**
Timur	5	2	**3**
Dilnoza	5	4	**1**
Sardor	2	3	**1**

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

$$= \frac{4+1+3+1+1}{5} = 2$$

Give the model parameters to learn

Let the data choose the weight of each mark.

name	statistics	python	ML mark	prediction
Aziz	5	4	5	4.45
Malika	4	5	4	4.73
Timur	5	5	5	5.10
Dilnoza	3	4	5	3.71
Sardor	3	3	2	3.06

$$\hat{y} = w_1\, x_{\text{stats}} + w_2\, x_{\text{python}}$$

$w_1 = 0.37,$ $w_2 = 0.65$ chosen to make the error on these students small

Compare models by their loss

Mean squared error on the five students. Lower is better.

5.60

a random mark

0.75

the average of the two marks

0.73

learned weights

Learning means minimising a loss

$$\hat{w} = \arg\min_{w} \frac{1}{n} \sum_{i=1}^{n} L\big(y_i,\ a_w(x_i)\big)$$

$a_w(x)$	the model; *w* are its parameters
$L(y, \hat{y}) = (y - \hat{y})^2$	the loss, here the squared error
n	the number of training objects

The goal is low loss on new data

0.73 is the error on the five students the weights were fitted on. What counts is the error on students the model has not seen. This is called generalisation.

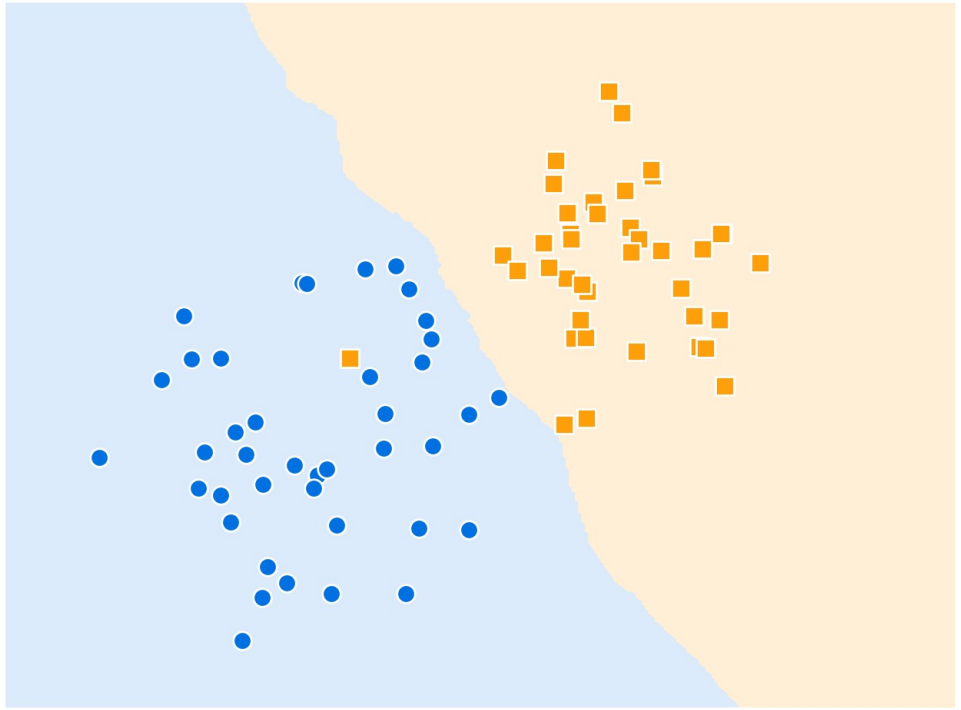
Main concepts

term	also called	notation
Dataset	sample; the features *X* and the target *y*	(X, y)
Object	observation, data point; a row of *X*	x_i
Feature	predictor, field; a column of *X*	x^j
Target	label, answer	y
Model	hypothesis, algorithm	$a(x)$
Prediction	what the model outputs	$\hat{y}$
Loss	error function	$L(y, \hat{y})$

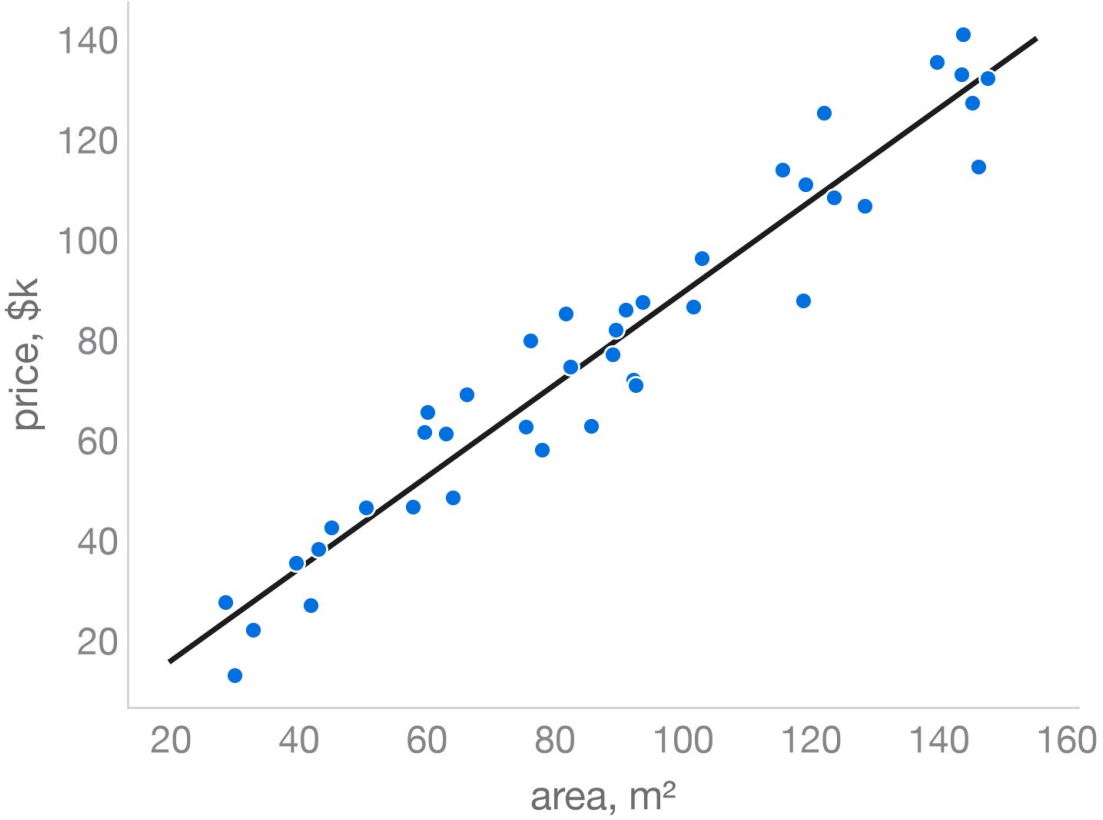
Types of learning

Supervised: objects with answers

Classification: y is a class



Regression: y is a number

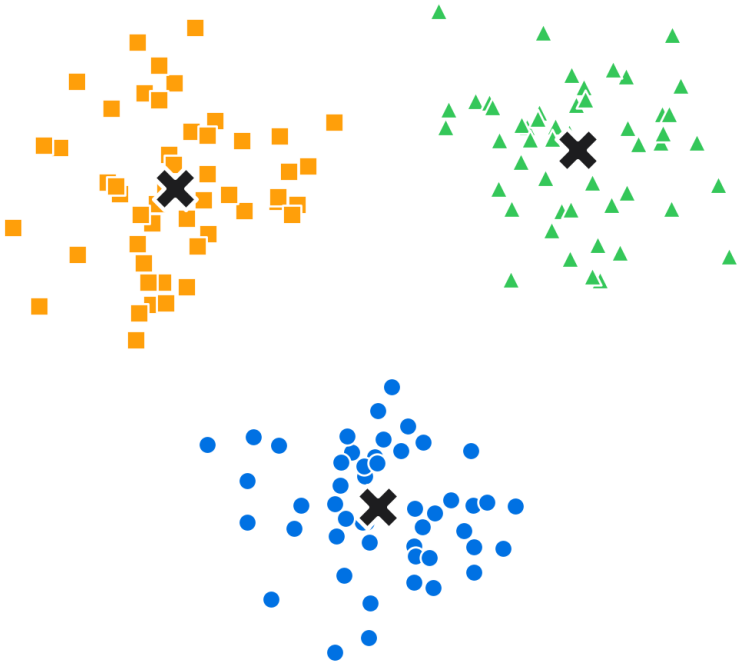


Unsupervised: find structure

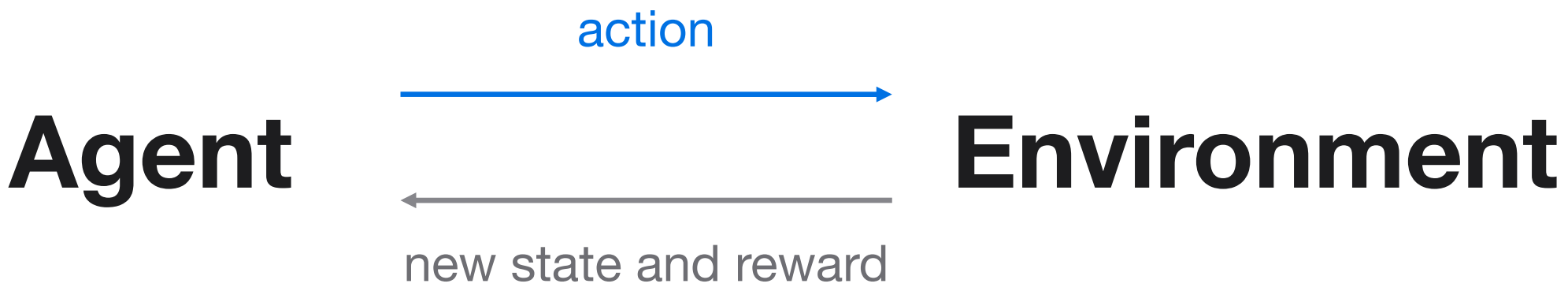
No answers, only objects



Three clusters found by k-means



Reinforcement: act, get a reward



Self-supervised: hide a part, predict it

Plov is the national dish of

Uzbekistan

Raw text supplies the answers for free. This is how large language models are pretrained.

Tasks we will meet in this course

Regression
a number: the price of a flat

Classification
a class: spam or not

Clustering
groups of similar objects: client segments

Dimensionality reduction
many features squeezed into a few

Anomaly detection
unusual objects: odd card payments

Generation
new text, images or sound

Which type of learning is it?

1. A spam filter trained on emails users marked
2. Flat prices in Tashkent from listings
3. Grouping bank clients into segments

Which type of learning is it?

4 A robot learning to walk

5 Pretraining a language model on raw text

6 Finding fraud without fraud labels

Datasets

Garbage in, garbage out

“On two occasions I have been asked, ‘Pray, Mr. Babbage, if you put into the machine wrong figures, will the right answers come out?’”

Charles Babbage, 1864

Where to find datasets

Kaggle

Datasets, notebooks and competitions

kaggle.com/datasets

Hugging Face

Text, images and audio, loaded in one line

huggingface.co/datasets

UCI repository

Classic tables, many from the 1990s and 2000s

archive.ics.uci.edu

Dataset Search

Google's search engine over dataset pages

datasetsearch.research.google.com

Competitions

Real tasks with a leaderboard

mlcontests.com

Croissant

One metadata format across these sites

mlcommons.org

Ask a chatbot to search for you

A deep-research request saves hours. Then check every link and licence yourself.

```
You are an ML researcher. Give an overview of open datasets
for speech recognition in Uzbek: size, licence, and where to
download each one. Include the datasets that recent papers
use.
```

Famous datasets

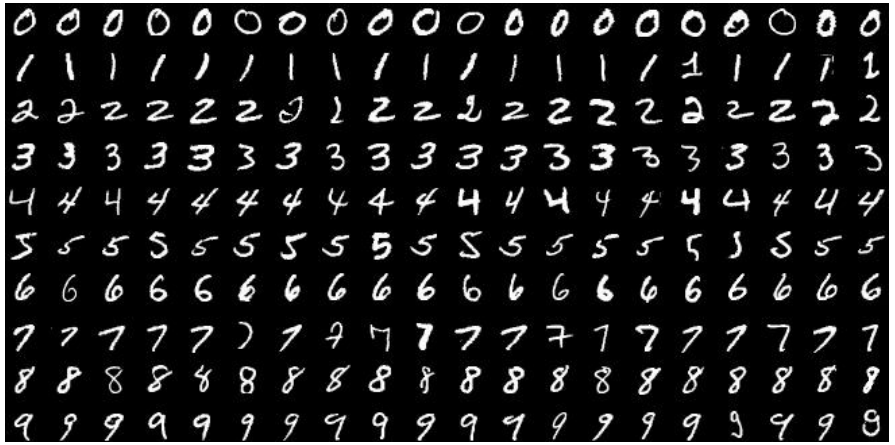
Iris, 1936

150 flowers, 4 measurements, 3 species



MNIST, 1998

70,000 handwritten digits, 28 × 28 pixels



ImageNet, 2009

14 million photos; its challenge used 1000 classes

Breast cancer

569 tumours, 30 features

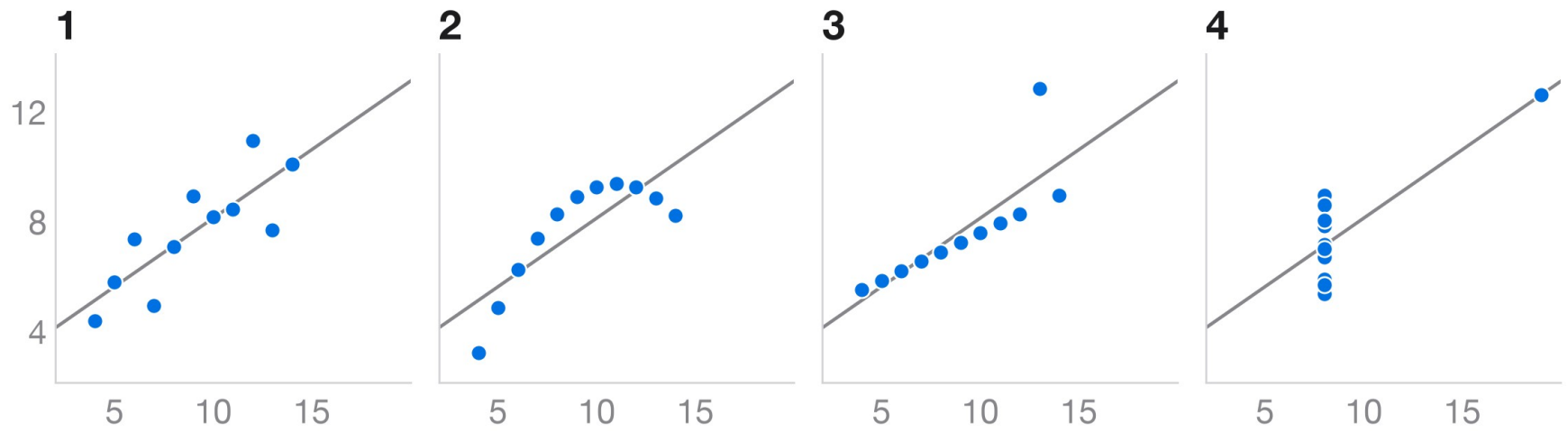
Boston house prices

removed from scikit-learn: one feature was built from race

Exploratory data analysis

Same statistics, different data

Four datasets with the same means, variances, correlation and line.



$$\bar{x} = 9$$

$$\bar{y} = 7.50$$

$$s_x^2 = 11$$

$$s_y^2 \approx 4.1$$

$$r_{xy} = 0.82$$

$$\hat{y} = 3 + 0.5\,x$$

Plot the data before you trust the numbers.

F. J. Anscombe, Graphs in Statistical Analysis, 1973

Steps of EDA

1. Understand the task
2. Load the data and look at it
3. Handle missing values
4. Explore each feature
5. Transform features
6. Look at relationships
7. Handle outliers
8. Write down what you found

After GeeksforGeeks, What is exploratory data analysis

EDA on our five students

```
df.shape          # (5, 8)
df.info()         # types, non-null
df.describe()     # mean, std, min, max
df.isna().sum() # eye colour: 1 NA
```

Eye colour has one missing value

Two features are categories: they need encoding

Each name is unique, so the name column tells us nothing

Five rows: every conclusion is shaky

One-hot encoding

Categories become numbers: one column per category, 1 where it matches.

name	eye colour	brown	blue	green
Aziz	brown	**1**	0	0
Malika	brown	**1**	0	0
Timur	blue	0	**1**	0
Dilnoza	green	0	0	**1**
Sardor	NA	0	0	0

Sardor's eye colour is missing, so all three columns are 0.

More on encodings: A. Dyakonov, feature engineering lecture; feaz-book.com

The ML workflow

Problem, data, model

Problem

What goes in, what comes out, and how we score it: a metric and a loss

Data

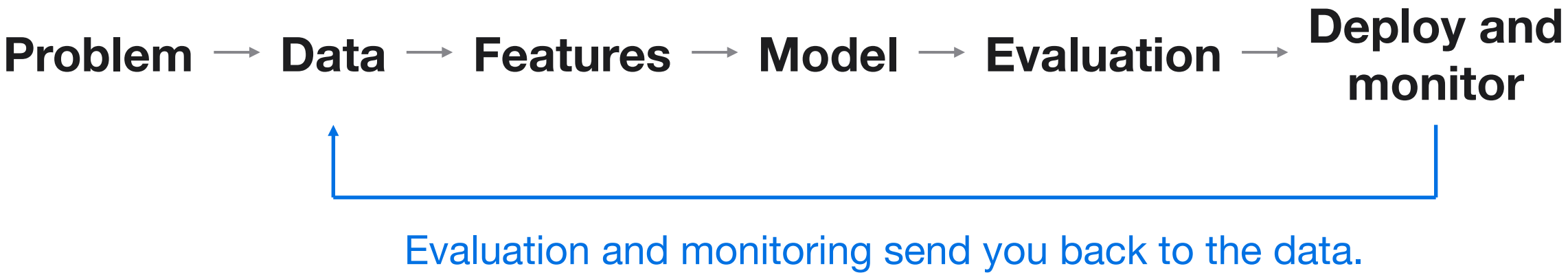
A sample from an unknown distribution, split into train and test

Model

Parameters it learns from data, hyperparameters we choose

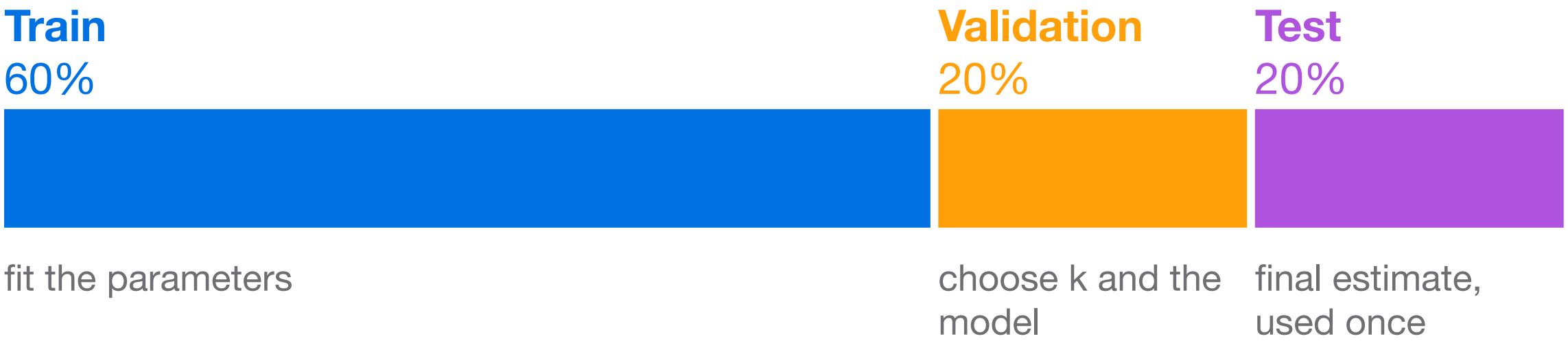
One problem, one dataset and one model make an experiment. Change any of them and you get a new one.

The ML workflow is a loop



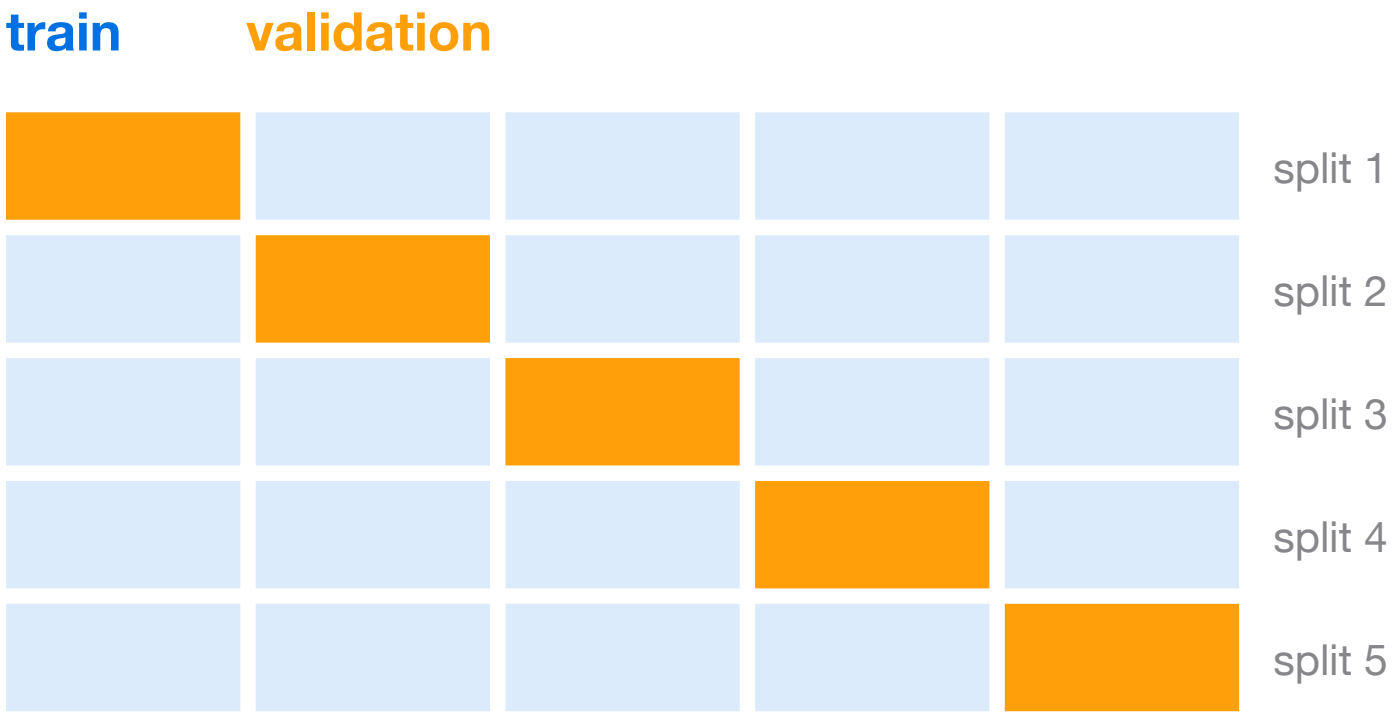
Most of the time goes into the problem, the data and the evaluation.

Train, validation, test



Predicting the future? Split by time.

Small data: cross-validation



Every object is in validation exactly once.

Average the five scores.

Leaks that inflate your score

1. A scaler fitted on all data before the split
2. The same object in train and in test
3. Choosing the model by its test score

Fit everything on the training part only

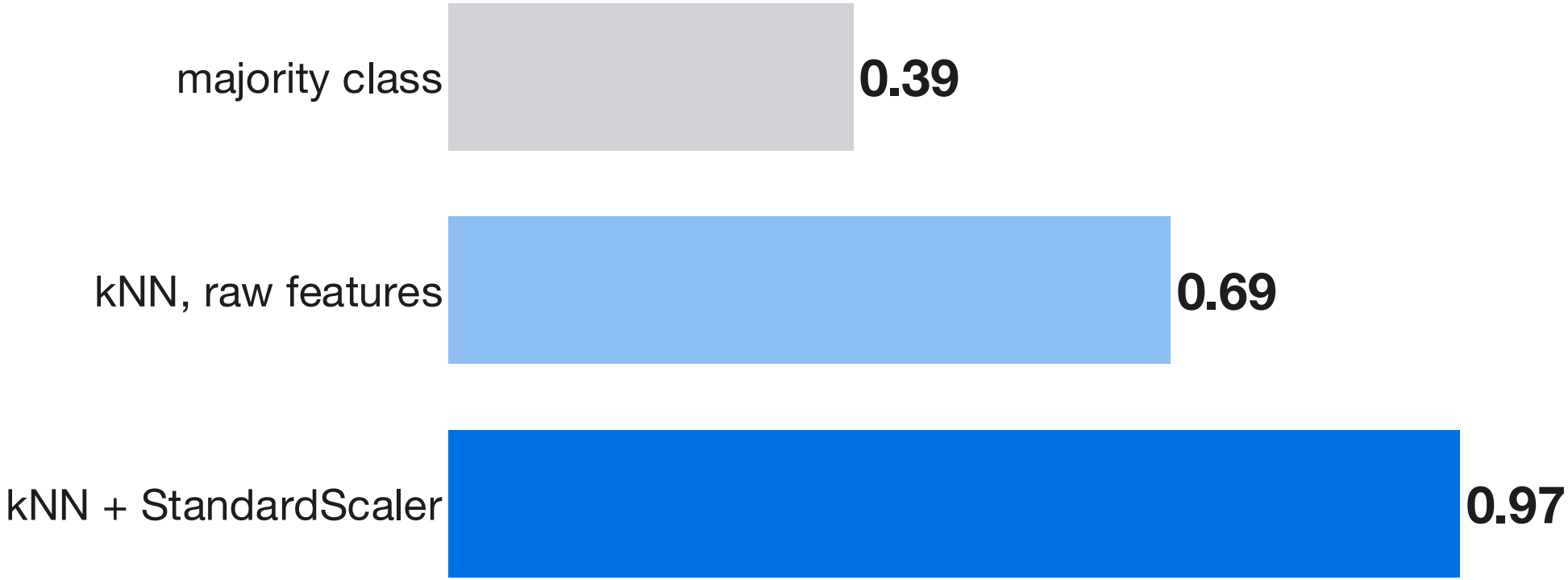
Scalers, encoders, feature selection, k. In scikit-learn, a Pipeline does it for you.

A benchmark is a test set too

Once it is public, it leaks into training data and people tune on it. That is why new benchmarks keep appearing.

Baseline first

Wine dataset, validation accuracy, k = 5



The Python ML stack

NumPy

arrays and linear algebra

pandas, Polars

tables and EDA

scikit-learn

classic ML, kNN today

matplotlib

plots

PyTorch

deep learning, later

Jupyter, Colab

seminars and homework

Store tables in Parquet

CSV

Plain text you can open anywhere

Big files, slow to read

No types: dates and IDs need parsing again

Parquet

Stores data by column, compressed

Fast to read, keeps the types

pandas and Polars read and write it

Log every run

run	model	k	scaler	val acc	commit
1	majority		no	0.39	`a1f3c2e`
2	kNN	5	no	0.69	`a1f3c2e`
3	kNN	5	standard	0.97	`b7d901a`
4	kNN	15	standard	1.00	`b7d901a`

Classic ML and foundation models

Foundation models

Pretrained on huge, broad data, mostly self-supervised

Adapted by prompting, fine-tuning, or used for embeddings

GPT, Claude, Gemini, Qwen, Llama; CLIP for images; Whisper for speech

Open models are close behind

4 months

Open-weight models trail the best closed ones by about four months.

Epoch AI, May 2026

46 vs 60

Best open model (Kimi K3.1 Pro) against the best closed one (Gemini 4 Carbon).

Artificial Analysis Intelligence Index, November 2026

Which one wins?

Classic ML

Tables with many labelled rows

Tight latency, cost or regulation

Runs on your servers or a phone

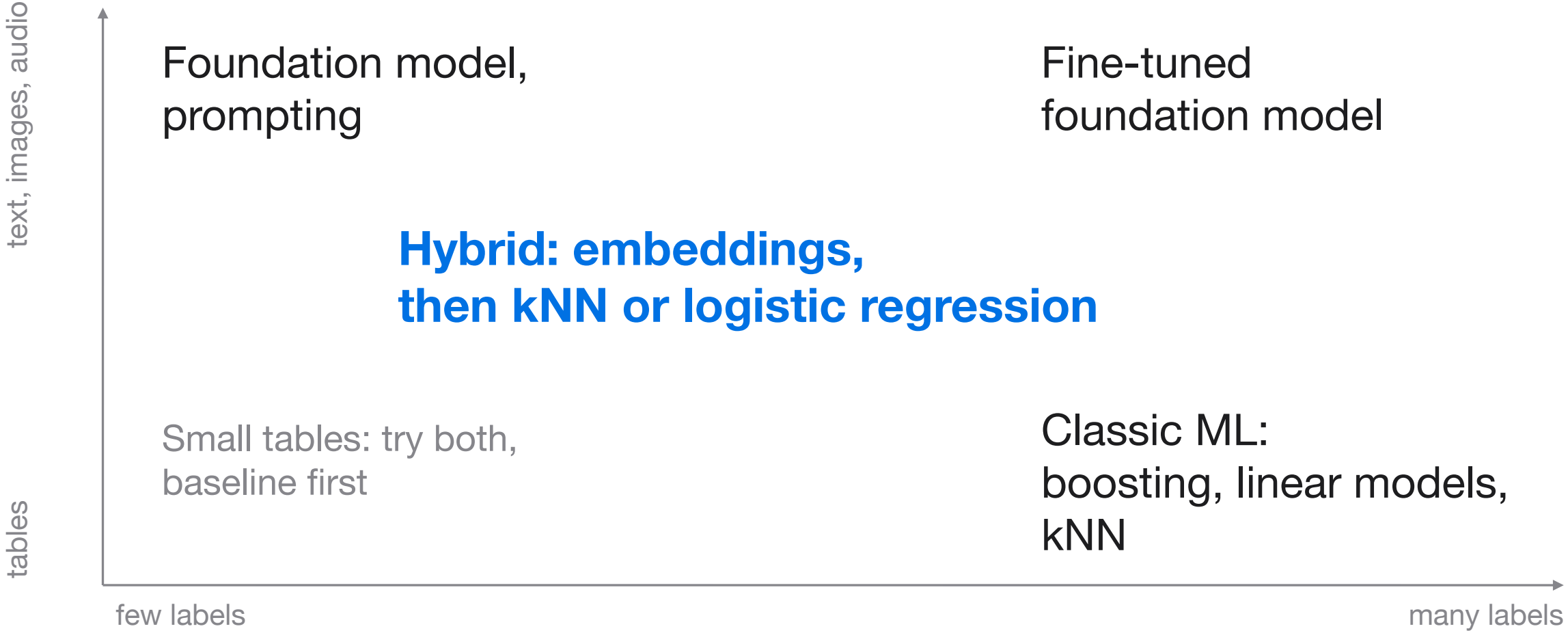
Foundation models

Text, images, audio

Few or no labels

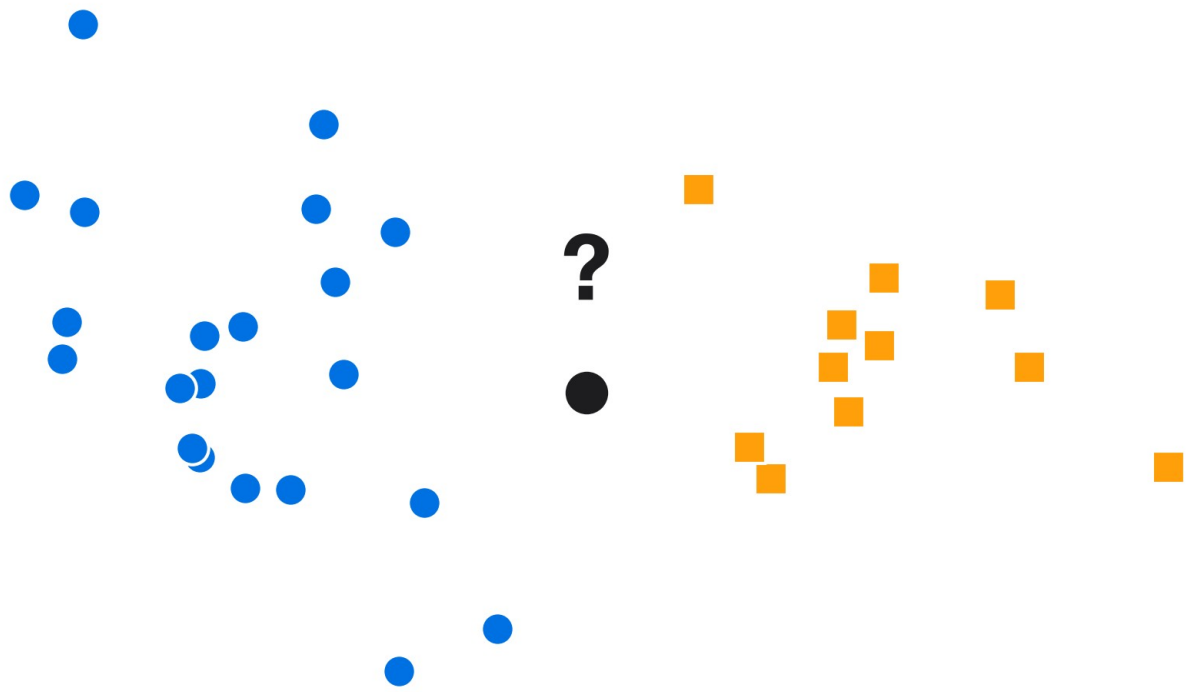
Generation, world knowledge, many languages

Hybrid: embed, then kNN



k nearest neighbours

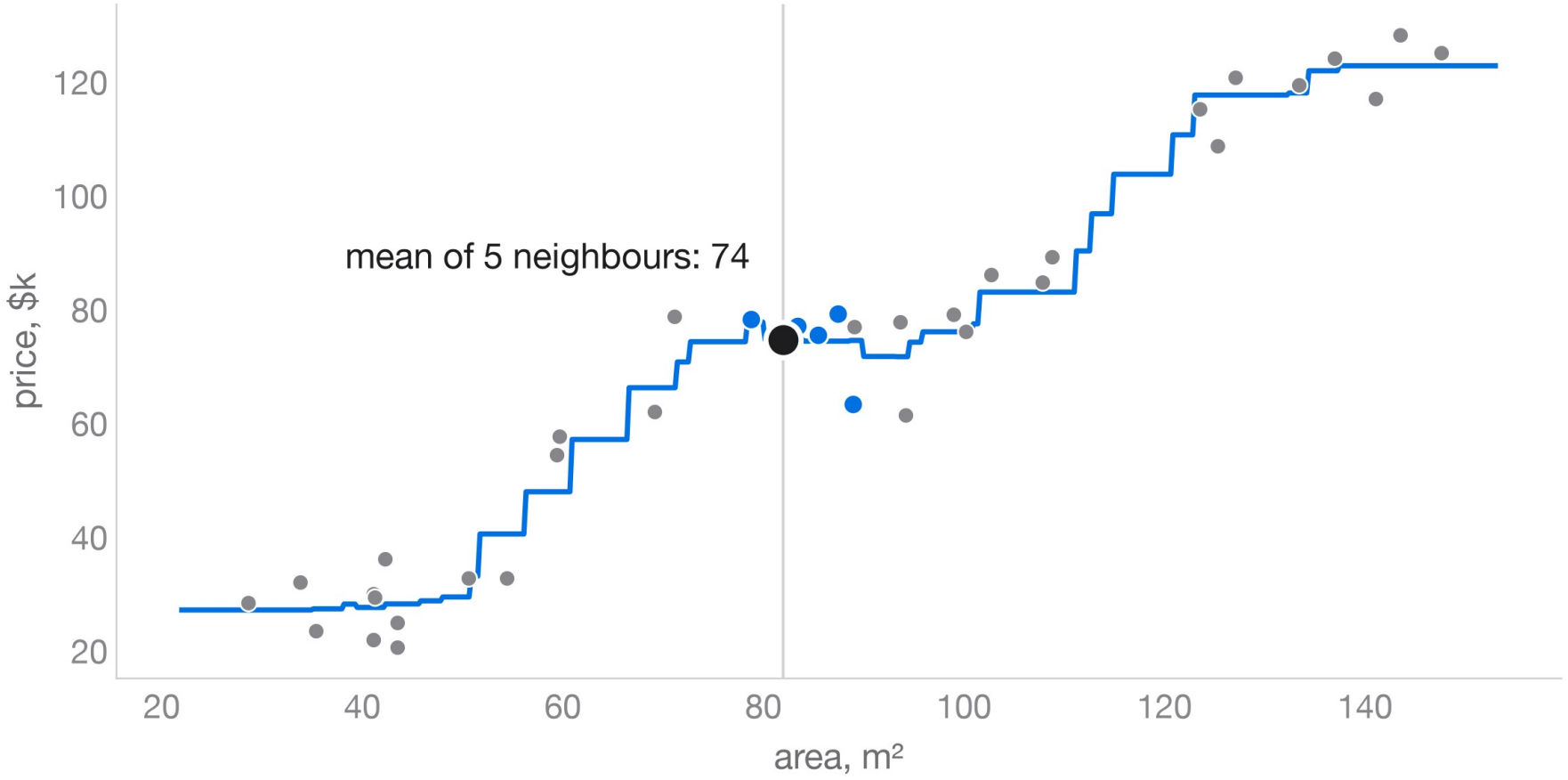
Which class is the new point?



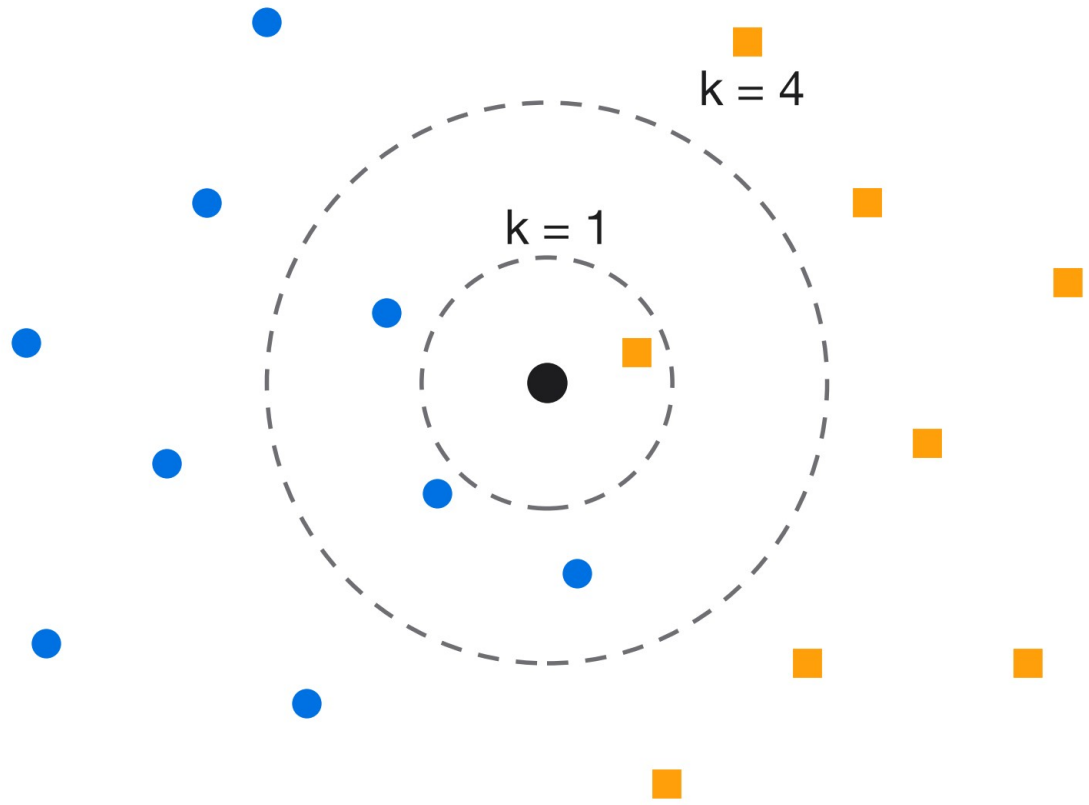
kNN in three steps

1. Measure the distance to every training object
2. Take the k closest
3. Vote for a class, or average the numbers

Regression: average the neighbours



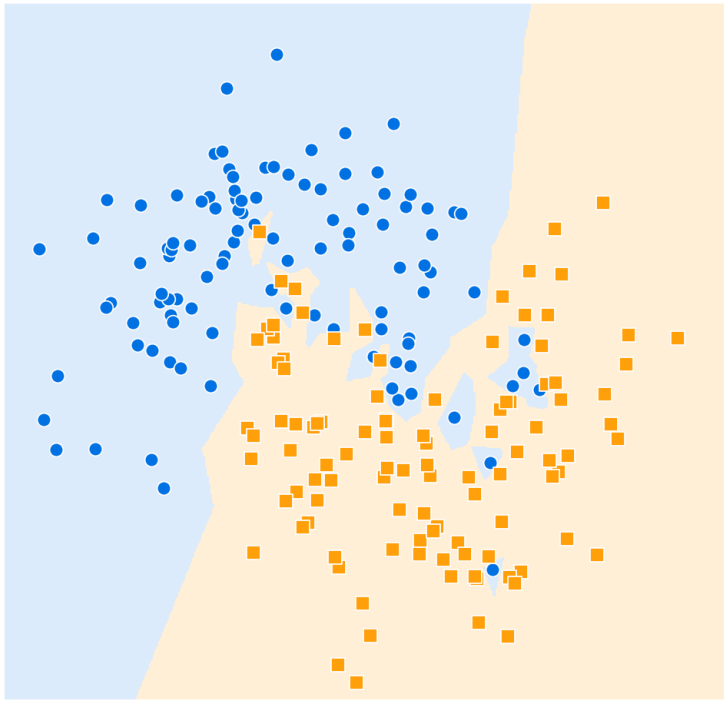
The answer depends on k



k = 1: square

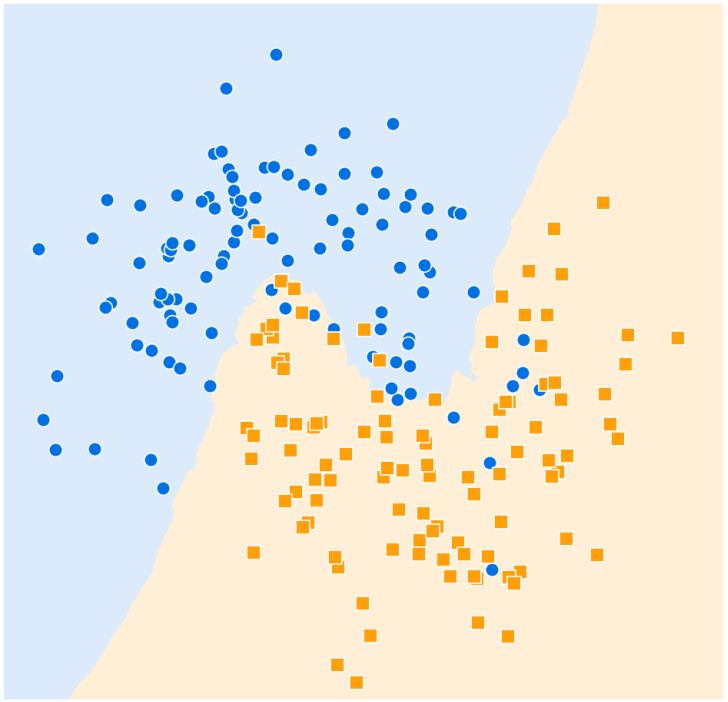
k = 4: circle

Small k overfits, large k underfits



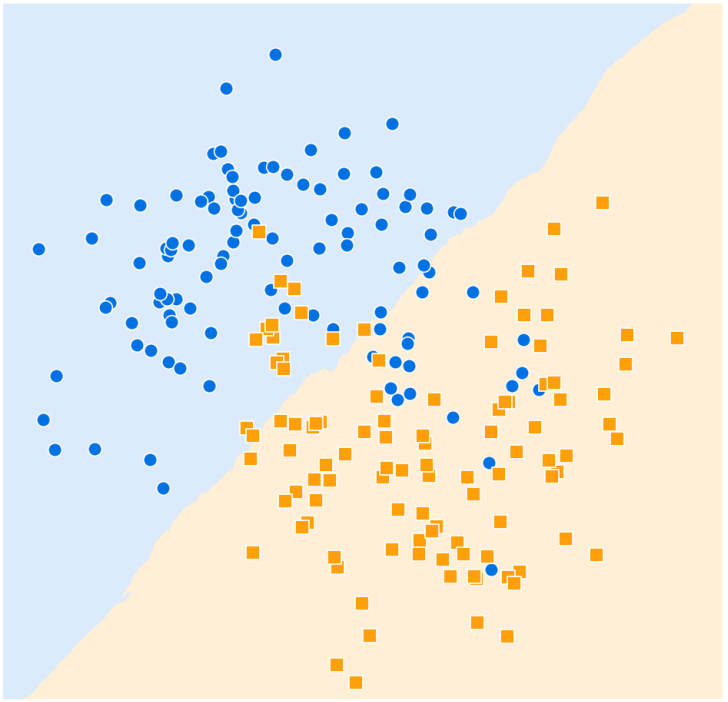
k = 1

train 1.00 test 0.90



k = 15

train 0.91 test 0.93

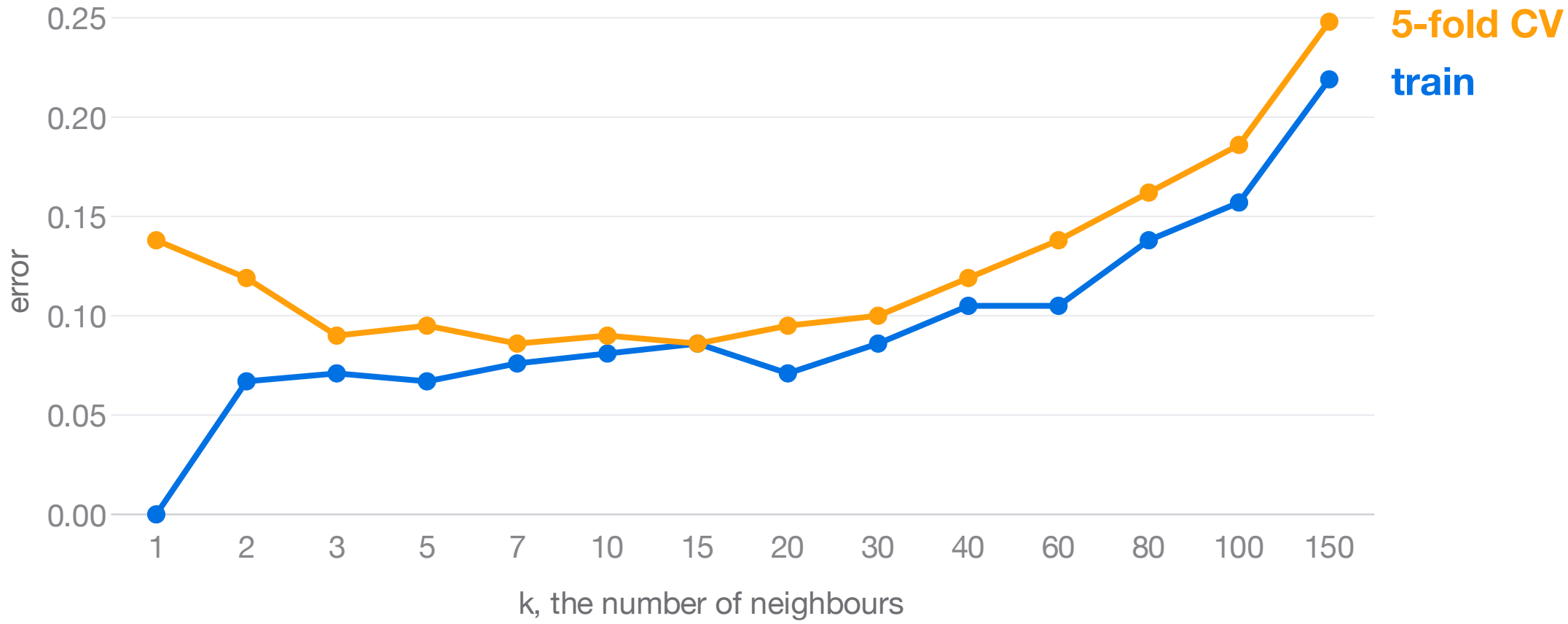


k = 100

train 0.84 test 0.76

Choose k by cross-validation

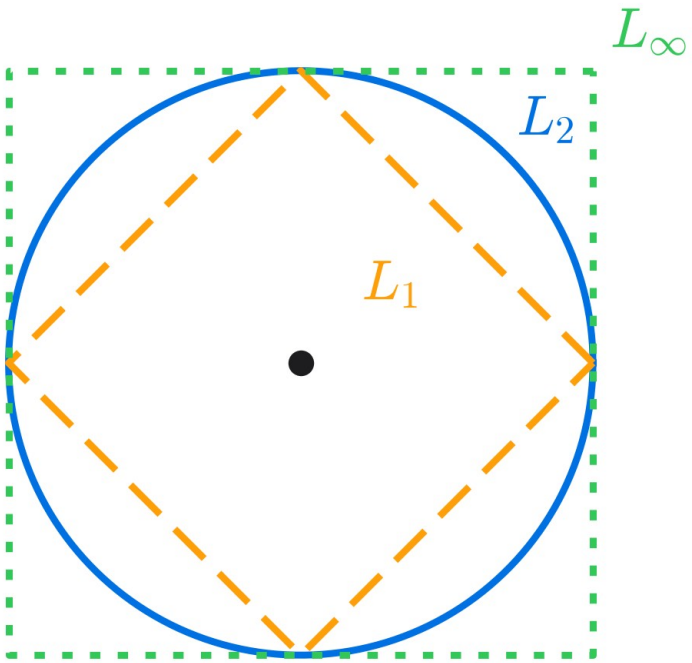
Noisy two moons, 210 training points



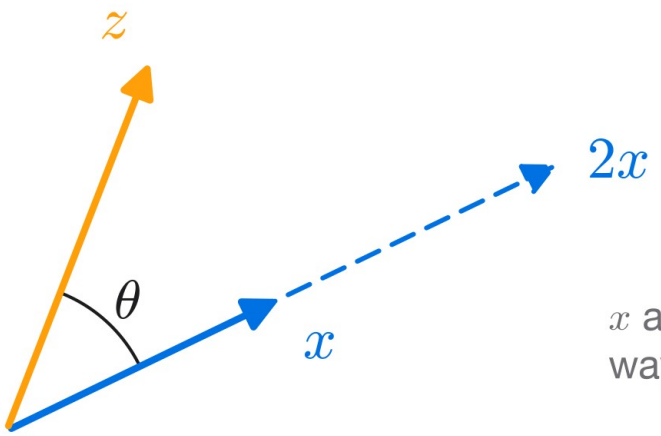
The metric defines “close”

Cosine for texts and embeddings. Hamming for strings and categories.

Points at distance 1

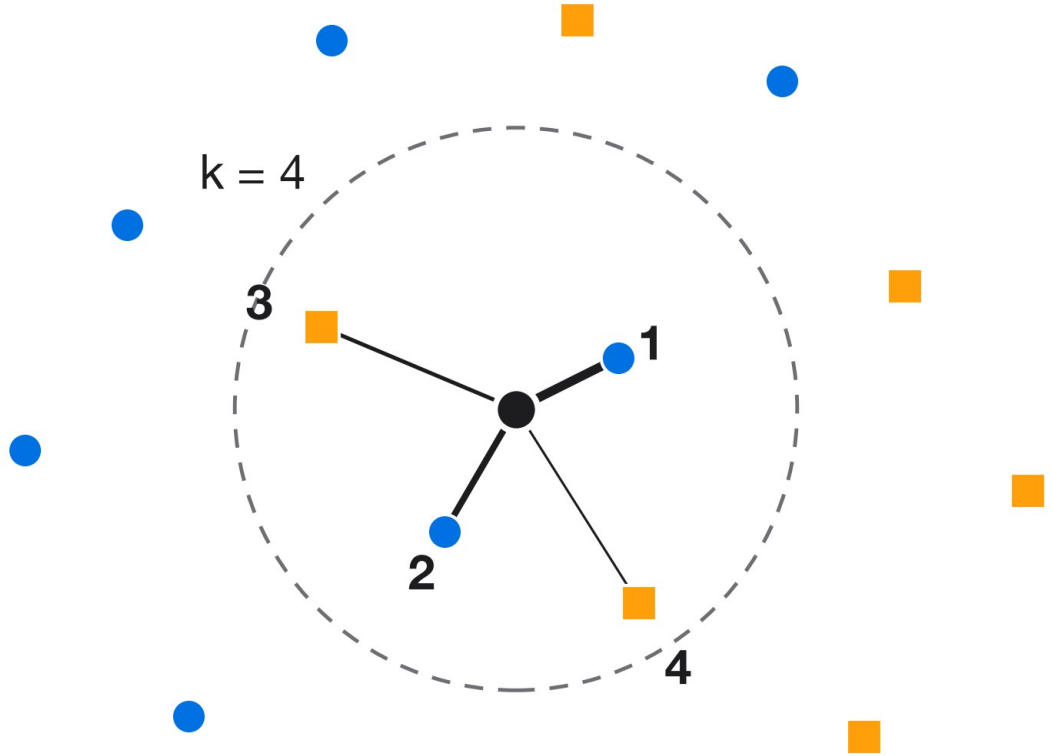


Cosine: only the angle counts



x and $2x$ point the same way: cosine distance 0

Closer neighbours get more weight



by rank

$$w_{(i)} = w_i$$

by distance

$$w_{(i)} = w\big(d(x, x_{(i)})\big)$$

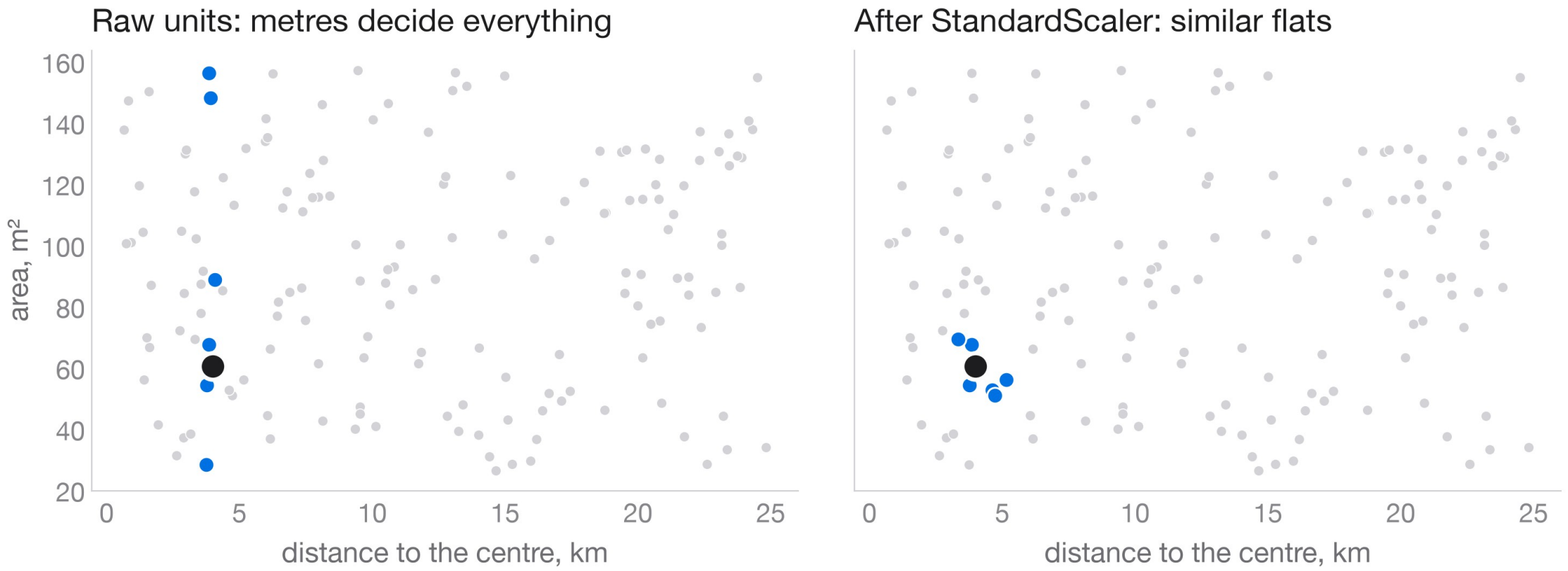
$$p(\text{circle}) = \frac{w_1 + w_2}{\sum_{i=1}^{4} w_i}$$

Without weights it is a 2 to 2 tie.

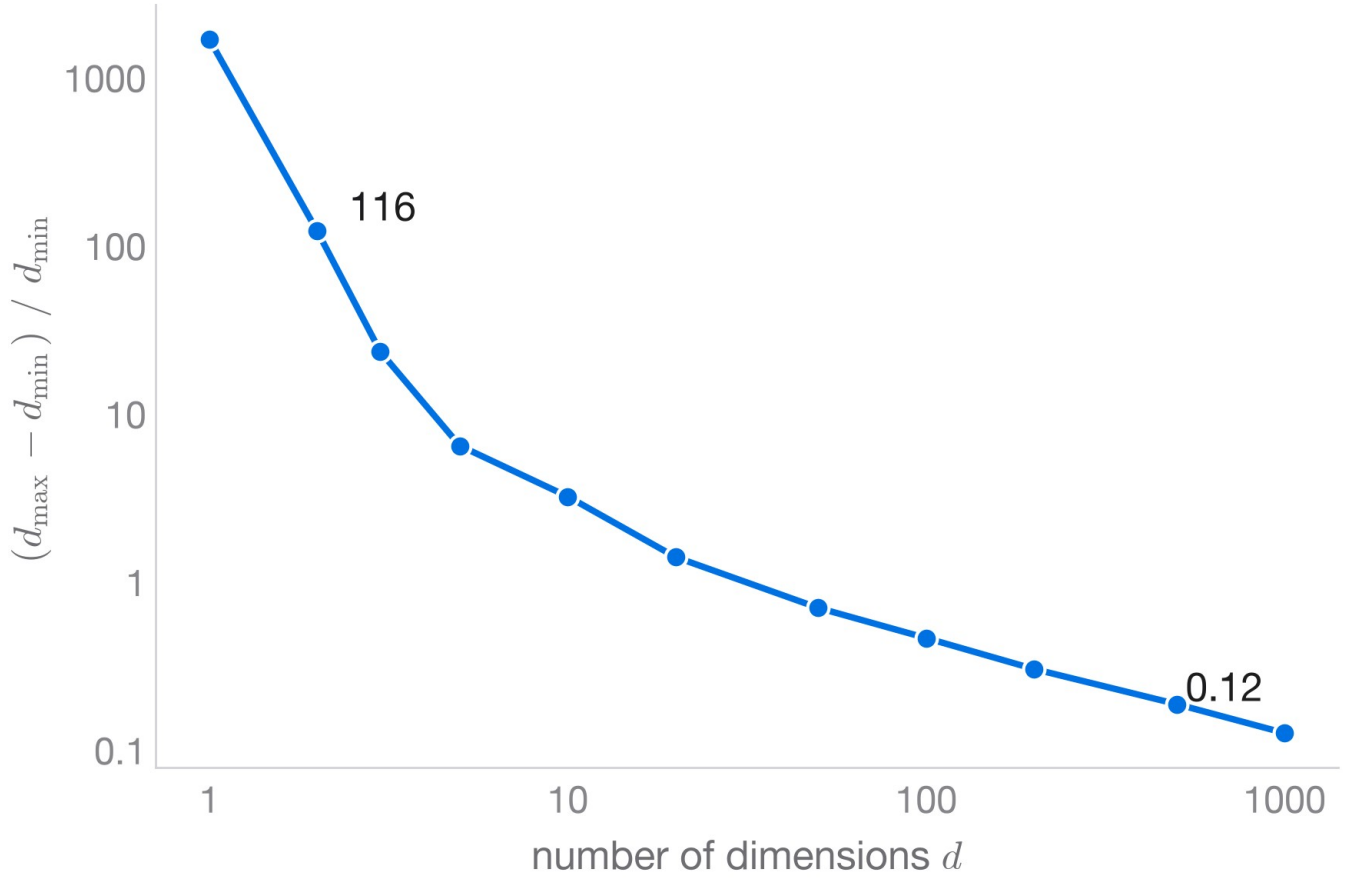
Scale the features

$$x' = \frac{x - \mu}{\sigma}$$

Flat prices: area in m², distance to the centre in metres.



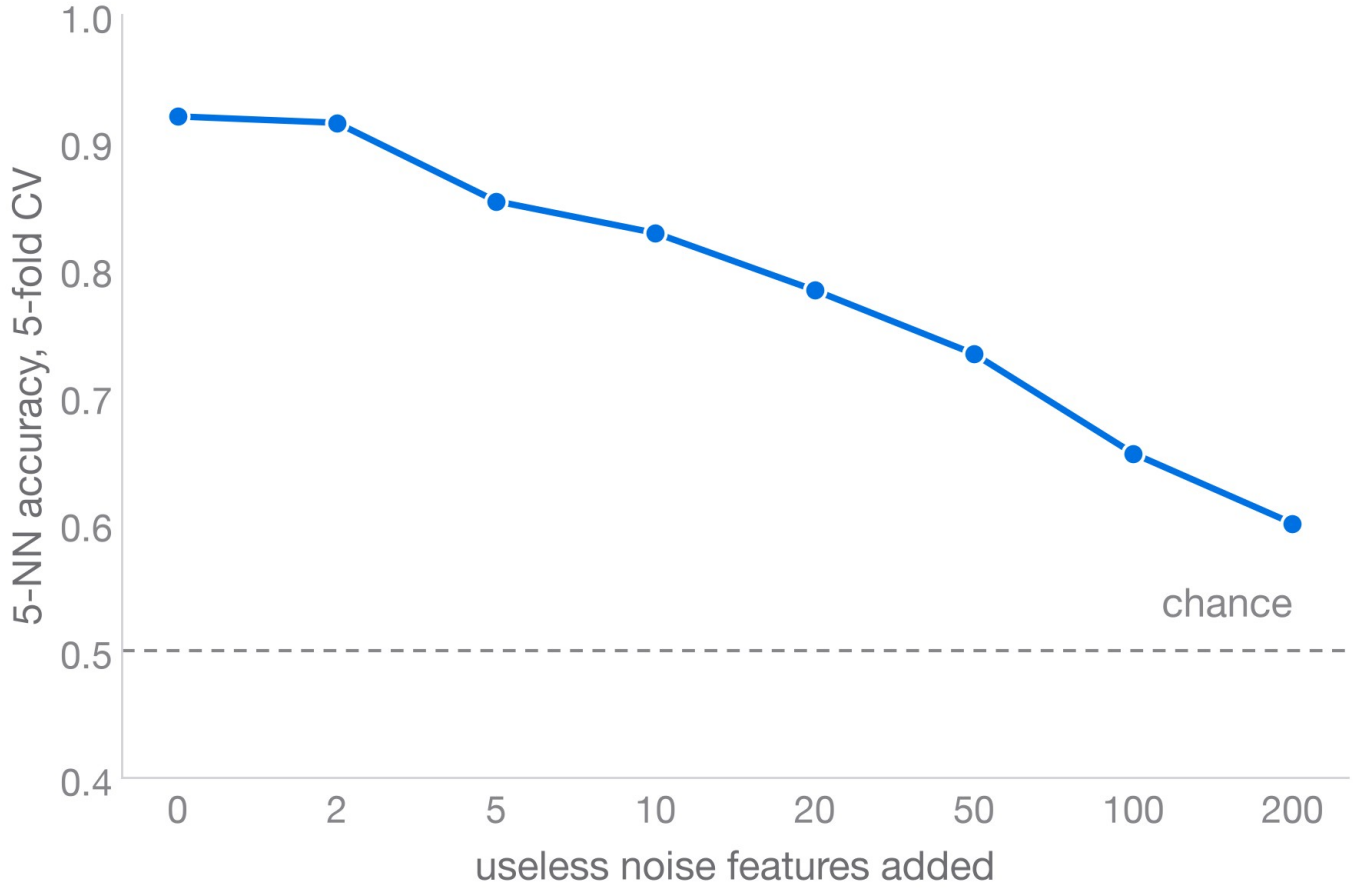
High dimensions: everything is far



0.12

In 1000 dimensions the farthest point is only 12% farther than the nearest.

Useless features hurt kNN



0.60

accuracy with 200 noise features, down from 0.92

0.1 s

to find 10 nearest neighbours by brute force

10^6 vectors of 100 numbers, 10^9 operations per second. Fine once, too slow for thousands of queries a second.

A kNN index makes search fast

Build it once, then search. Approximate search trades a little accuracy for a lot of speed.

Trees

k-d tree and ball tree, in scikit-learn

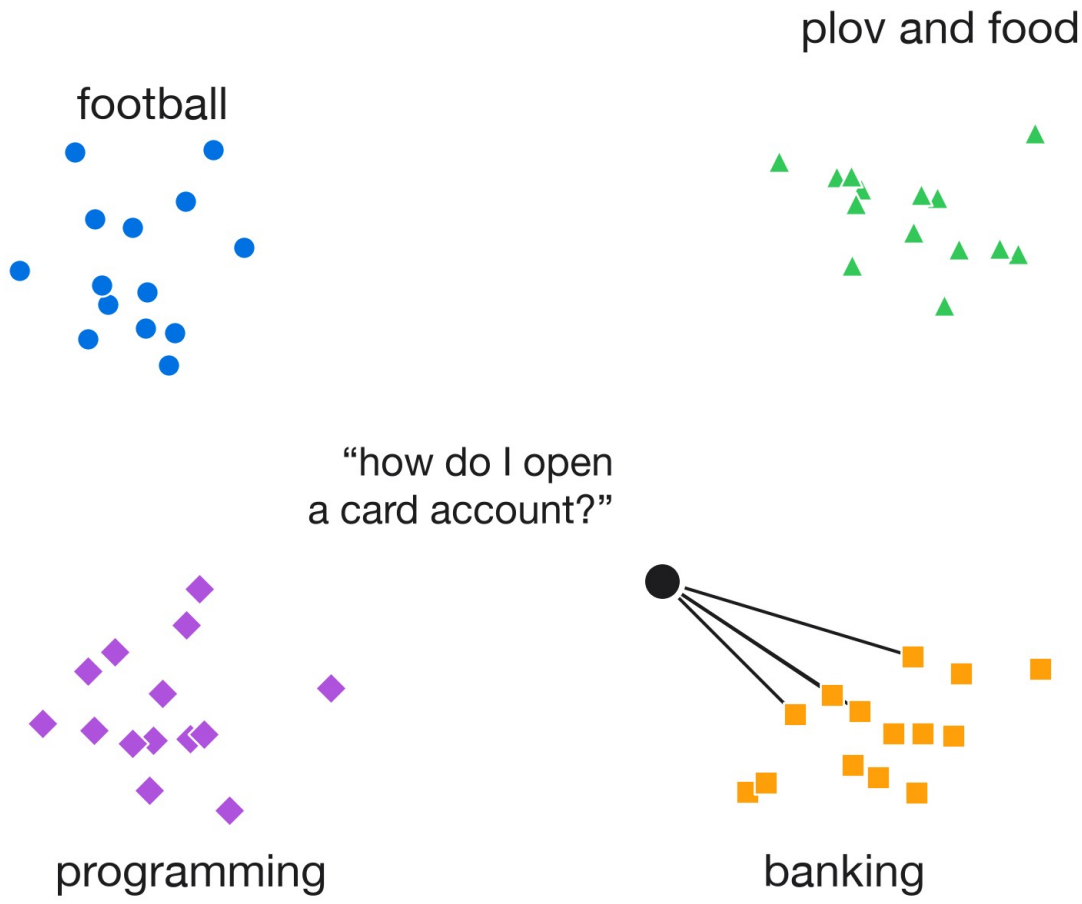
Clusters

IVF: search only the nearest clusters

Graphs

HNSW, the usual choice today

kNN today: search in embedding space



Semantic search and RAG

Recommendations

Face recognition and deduplication

kNN: pros and cons

Pros

No training: just add data

Classes and numbers alike

Easy to explain: “because of these neighbours”

Cons

Slow search without an index

Needs scaled, relevant features

Weak in high dimensions

Revise

ML learns rules from data. We judge it on new data.

Rows are objects, columns are features. Look at the data first.

Split train, validation, test. No leaks. Baseline first.

kNN: pick k and the metric by validation. Scale features.

Exit quiz

Scan the code.

Enter your full name. One question at a time, 30 seconds each.

If time allows, we then play the multiplayer game Neighbours.



165.22.93.84

Practice and homework

Seminar

Two notebooks: workflow and baseline, then kNN practice

Interactive

Nine pages at 165.22.93.84/viz: play with k, metrics and scaling

Homework 1

kNN from scratch in NumPy: two loops, one loop, no loops

Links to the notebooks and the homework are in the course chat.

To read

Books

Bishop, Pattern Recognition and Machine Learning

Hastie, Tibshirani, Friedman, The Elements of Statistical Learning

Murphy, Machine Learning: A Probabilistic Perspective

Online

Yandex ML handbook, in Russian: education.yandex.ru/handbook/ml

mlcourse.ai, in English and Russian

StatQuest videos on YouTube

Next time: maximum likelihood and linear regression

Bring your questions about homework 1.

Questions?

Makhmudjon Sodikov, YMLS Uzbekistan

YMLS



girafe ai